

1. Suppose (x_n) is a real sequence and $x_n \rightarrow \ell > 0$. Show that there exists N so that if $n \geq N$ then $x_n > \ell/2$. Then show $1/x_n \rightarrow 1/\ell$. Note that the sequence $1/x_n$ is perhaps only defined for $n \geq N$ to avoid division by zero.
2. Suppose $\Omega \subset \mathbb{R}$, that a is a limit point of Ω , and that $f : \Omega \rightarrow \mathbb{R}$. Show that $\lim_{x \rightarrow a} f(x) = \ell$ if and only if whenever (x_n) is a sequence in Ω converging to a , $f(x_n) \rightarrow \ell$. You are encouraged to modify the similar proof seen in class on the sequential characterization of continuity.
3. Sutherland 4.13
4. Sutherland 4.14 (AI Free)
5. Sutherland 4.15
6. Use Definition 4.21 and no more advanced tools to directly show

$$\lim_{x \rightarrow 1} \frac{1}{x^2 + 1} = \frac{1}{5}.$$

That is, start your proof with "Let $\epsilon > 0$ ", exhibit a candidate δ , and show that your candidate works.

7. For $x, y \in \mathbb{R}$ define $d(x, y) = |y - x|$. Show that a sequence of real numbers (x_n) converges to a limit ℓ if and only if $d(x_n, \ell) \rightarrow 0$.
8. Sutherland 5.1
9. AI Use. Please describe specifically how you used AI to help complete this assignment.